**Code No: 114CW** **R13**

**JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD**
**B. Tech II Year II Semester Examinations, September/October - 2023**
**ENVIRONMENTAL STUDIES**
**(Common to CE, ECE, CSE)**

**Time: 3 Hours** **Max. Marks: 75**

**Note:** i) Question paper consists of Part A, Part B.
ii) Part A is compulsory, which carries 25 marks. In Part A, answer all questions.
iii) In Part B, Answer any one question from each unit. Each question carries 10 marks and may have a, b as sub questions.

**PART - A**

**(25 Marks)**

| | | |
|---|---|---|
| 1.a) | Write down ecosystem services. | [2] |
| b) | Define food chains. | [3] |
| c) | Define Groundwater. | [2] |
| d) | Write down various sources of surface water. | [3] |
| e) | Define ecosystem diversity. | [2] |
| f) | Write down threats to Biodiversity. | [3] |
| g) | Define water pollution. | [2] |
| h) | Define noise pollution. | [3] |
| i) | Define Environmental Management Plan. | [2] |
| j) | Explain Sustainable Development Goal no.6. | [3] |

**PART - B**

**(50 Marks)**

| | | |
|---|---|---|
| 2. | Explain the following:<br>a) Biogeochemical cycles.<br>b) Ecological pyramids. | [5+5] |
| | **OR** | |
| 3.a) | Define ecosystem along with scope and importance. | |
| b) | Explain classification and structure of an ecosystem. | [5+5] |
| 4.a) | Define floods and write down various causes for floods. | |
| b) | Explain Forest Resources and Energy Resources. | [5+5] |
| | **OR** | |
| 5.a) | Explain about Land resources. | |
| b) | Compare renewable and non-renewable energy sources. | [5+5] |
| 6.a) | Discuss the characteristics of mega diversity nation with examples. | |
| b) | Why man- wild life conflicts are occurring? | [5+5] |
| | **OR** | |
| 7.a) | When do you prefer in-situ and ex-situ conservation? Discuss. | |
| b) | Write down the salient features of National Biodiversity act. | [5+5] |

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8.a) What are the health hazards due to noise pollution? Discuss.
b) Elaborate primary and secondary wastewater treatment methods. [5+5]

**OR**

9.a) Write down the drinking water quality standards in India.
b) List and explain various sources of Air pollution. [5+5]

10.a) Write down the salient features of Air Act.
b) Elaborate biomedical waste Management and handling rules. [5+5]

**OR**

11.a) Explain life cycle assessment and Environmental ethics.
b) Write down the salient features of Wild life Act. [5+5]

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